

Capstone Lightning Talk #1

Three Potential Project Proposals

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TOPIC 1

Loan Status Prediction

Data Source: Kaggle — Financial Loan Dataset

PROBLEM STATEMENT

Can we predict the status of a loan using the borrower's financial information, employment details, loan characteristics, and repayment history?

GOAL / AIM

Help lenders flag loans that are likely to default early, so risk can be managed more proactively.

Loan Status Prediction



PROBLEM TYPE

Multiclass Classification— Target: loan_status (Fully Paid / Charged Off / Current)



MISSING VALUES

1,438 missing in emp_title



FEATURE TYPES

15 categorical, 9 numerical columns



OUTLIERS

Outliers were found in all numerical features.
The largest values appeared in annual_income (6M) and total_payment (58.6K).

TOPIC 2

American Companies Bankruptcy Prediction

Data Source: Kaggle — American Companies Bankruptcy Prediction

PROBLEM STATEMENT

Predict whether a company is likely to go bankrupt based on its financial indicators.

GOAL / AIM

Help investors, banks, and financial institutions flag high-risk companies before making decisions.

American Companies Bankruptcy Prediction



PROBLEM TYPE

Classification (binary) — target: status_label (alive / failed)



MISSING VALUES

No missing values found.



FEATURE TYPES

2 categorical, 19 numerical columns



OUTLIERS

Outliers were found in most financial features.
The largest values appeared in retained_earnings and net_income.
These values were kept because they likely represent real companies rather than data errors.

TOPIC 3

Credit Card Default Prediction

Data Source: Kaggle — Default of Credit Card Clients Dataset

PROBLEM STATEMENT

Can we predict whether a credit card customer will default on their next payment using demographics, credit limit, repayment history, and bill/payment amounts?

GOAL / AIM

Help card issuers anticipate defaults and manage credit risk before it happens.

Credit Card Default Prediction



PROBLEM TYPE

Classification (binary) — Target: default.payment.next.month



MISSING VALUES

No missing values found.



FEATURE TYPES

9 categorical, 15 numerical columns



OUTLIERS

Outliers were found in most financial features.
The largest values appeared in PAY_AMT4 and PAY_AMT6.
These values were kept because they may represent genuine customer payment behavior.